

House Price Prediction

Objective:

To develop and evaluate machine learning models for predicting house prices based on property characteristics and amenities.

Dataset Overview:

- Dataset: Housing.csv
- Number of Rows: 545
- Number of Columns: 13
- Target Variable: Price
- Total Features: 12

The dataset contains information about residential properties, including features such as area, number of bedrooms, bathrooms, stories, parking spaces, furnishing status, and other amenities. The target variable is house price, which is predicted using machine learning regression models.

Models Used:

- Linear Regression
- Random Forest Regressor

Results:

Model	MAE	RMSE	R ² Score
Linear Regression	970,043	1,324,507	0.653
Random Forest Regressor	1,021,546	1,400,566	0.612

Key Findings:

- Area was the most influential feature affecting house prices.
- Bathrooms, stories, and parking facilities also contributed significantly.
- Linear Regression outperformed Random Forest with a higher R² score (0.653) and lower prediction errors.
- Location and amenities such as air conditioning and preferred area had a noticeable impact on house prices.

Conclusion:

- Successfully developed and evaluated machine learning models for house price prediction.
- Linear Regression achieved the best performance with an R² score of 0.653.
- Area was identified as the most influential feature affecting house prices.
- Bathrooms, stories, parking facilities, and other amenities also contributed significantly.
- Machine learning can effectively support real estate price estimation and decision-making.